Lab 2: NFT Smart Contract & Ownership Economy

实验二：NFT 智能合约与所有权经济

1. Objectives (实验目标)

1. **Master ERC-721 Standard**: Understand the industry standard for non-fungible tokens. (掌握 ERC-721 标准：理解非同质化代币的行业标准。)
2. **Understand Digital Scarcity**: Explore how code enforces uniqueness and limited supply. (理解数字稀缺性：探索代码如何强制执行唯一性和限量供应。)
3. **Advanced Data Mapping**: Deepen understanding of mapping(uint256 => address) for tracking ownership. (进阶数据映射：深化对 mapping 追踪资产所有权的理解。)
4. **IPFS & Metadata**: Learn how blockchain links to off-chain assets (images/art). (IPFS 与元数据：学习区块链如何链接链外资产。)

2. Background: The Digital Art Gallery (实验背景：数字艺术画廊)

In Lab 1, you built a transparent ledger for fungible ETH. However, some assets are unique (like a piece of art or a real estate title). **NFTs (Non-Fungible Tokens)** allow us to represent these unique assets on-chain. In this lab, you will create a "Digital Art Gallery" where users can mint their own unique masterpieces.

(在实验一中，你为可替代的 ETH 建立了一个透明账本。然而，某些资产是唯一的（如艺术品）。NFT 允许我们在链上表示这些唯一资产。本次实验你将创建一个“数字画廊”。)

3. Requirements (实验要求)

3.1 Basic Functionalities (基础功能 - 必做)

Write a contract named SimpleNFT using the OpenZeppelin library. (使用 OpenZeppelin 库编写名为 SimpleNFT 的合约。)

- **3.1.1. Minting Mechanism (铸造机制)**: Implement a mint() function that allows an EOA to create a new NFT. (实现 mint() 函数，允许外部账户创建新 NFT。)
- **3.1.2. Ownership Tracking (所有权追踪)**: Each NFT must have a unique tokenId. Use a counter to increment IDs. (每个 NFT 必须有唯一 ID，使用计数器累加。)
- **3.1.3. Metadata Linking (元数据链接)**: Assign a URI (Uniform Resource Identifier) to each token to point to its image. (为每个代币分配 URI 以指向其图像。)

3.2 Advanced Features (进阶功能 - 选做)

- **3.2.1. Minting Fee (铸造费)**: Charge a small amount of ETH (e.g., 0.01 ETH) for each minting action. (每次铸造收取少量 ETH 作为手续费。)
- **3.2.2. Supply Limit (总量限制)**: Set a MAX_SUPPLY (e.g., 100). The mint() function must revert once reached. (设置总量限制，达到后停止铸造。)
- **3.2.3. Royalties (版权费)**: Implement a logic where a percentage of future sales goes back to the creator. (实现版税逻辑，使未来的交易分成回流给创作者。)

4. Technical Concepts (核心概念)

- **ERC-721**: The "Interface" that defines how NFTs move between addresses. (定义 NFT 如何在地址间转移的接口。)
- **Token ID**: The unique "Serial Number" of the asset. (资产的唯一“序列号”。)
- **Burnable**: A function that allows an owner to "destroy" their NFT, removing it from the state. (允许所有者“销毁”其 NFT 的功能。)

5. Suggested Steps (建议步骤)

1. **Import OpenZeppelin**: Use @openzeppelin/contracts/token/ERC721/extensions/ERC721URIStorage.sol. (导入 OpenZeppelin 官方库。)
2. **Inherit & Construct**: Inherit the standard and set the Name and Symbol (e.g., "MyArt", "MART"). (继承标准并设置名称与缩写。)
3. **Deploy on Remix VM**: Test minting multiple tokens from different accounts. (在 Remix 虚拟机部署并测试从不同账户铸造。)

4. **Verify Ownership**: Use ownerOf(tokenId) to check which address owns which art piece. (使用 ownerOf 检查所有权归属。)

6. Assessment (实验考核)

Submit a report including: (提交包含以下内容的报告：)

1. **Remix Screenshot** showing the successful minting transaction and the tokenId in the logs. (展示铸造成功交易和日志 ID 的截图。)
2. **Ownership Check**: A screenshot of calling ownerOf for your minted NFT. (调用 ownerOf 确认所有权的截图。)